Hardy-Dominant Annular Coefficient Conversion and Its Sharp Gap Order

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Abstract

The direct H^∞ coefficient estimate of RH-300 used Cauchy's bound and has gap constant of order η^{-1} when $\rho = Re^\eta$. Since normalized H^∞ embeds contractively into H^2, the Hardy estimate improves this to order $\eta^{-1/2}$. We prove the Hardy constant exactly and show that its square-root order is optimal: equality is attained in H^2, while dyadic Rudin–Shapiro blocks give the same order on the H^∞ unit ball. At $R = 1.4$, $\rho = 1.41$, the constants are $139.007\ldots$ and $8.292\ldots$. No norm decay for the actual mismatch is asserted.

1 Coefficient functional

Let $f(z) = \sum_{n\geq 2} b_n z^n$ be analytic on $|z| < \rho$, put $x = R/\rho < 1$, and define

$$P_R(f) = \sum_{n\geq 2} |b_n| R^n.$$

Boundary H^2 norms use normalized angular measure.

2 Hardy-dominant theorem

Theorem 2.1. *For every $f \in H^2(\rho)$,*

$$P_R(f) \leq \|f\|_{H^2(\rho)} \frac{x^2}{\sqrt{1-x^2}}.$$

In particular, for $f \in H^\infty(\rho)$,

$$P_R(f) \leq \|f\|_{H^\infty(\rho)} \frac{x^2}{\sqrt{1-x^2}}.$$

The H^2 constant is exact. If $\rho = Re^\eta$, then the best uniform order on the H^∞ unit ball is also $\Theta(\eta^{-1/2})$ as $\eta \downarrow 0$.

Proof. Write $u_n = |b_n|\rho^n$. Cauchy–Schwarz gives

$$P_R(f) = \sum_{n\geq 2} u_n x^n \leq \left(\sum_{n\geq 2} u_n^2\right)^{1/2} \left(\sum_{n\geq 2} x^{2n}\right)^{1/2},$$

which is the first estimate. The second uses $\|f\|_{H^2} \leq \|f\|_{H^\infty}$. Equality in the Hilbert-space estimate is obtained by taking the boundary-scaled coefficient vector proportional to $(x^n)_{n\geq 2}$.

For the H^∞ lower order, let $N = 2^m$ and choose Rudin–Shapiro signs $\varepsilon_0, \dots, \varepsilon_{N-1}$ with

$$\sup_{|w|=1} \left| \sum_{j=0}^{N-1} \varepsilon_j w^j \right| \le \sqrt{2N}.$$

Then

$$f_N(z) = \frac{1}{\sqrt{2N}} \sum_{j=0}^{N-1} \varepsilon_j (z/\rho)^{j+2}$$

has $H^\infty(\rho)$ norm at most one and

$$P_R(f_N) = \frac{x^2(1-x^N)}{\sqrt{2N}(1-x)}.$$

Choose a dyadic N with $(2\eta)^{-1} \le N \le \eta^{-1}$. Since $x = e^{-\eta}$, the last expression is bounded below by $c\eta^{-1/2}$. The Hardy upper bound is asymptotic to $(2\eta)^{-1/2}$, proving the order. $\square$

3 Constants and protocol

At $R = 1.4$, $\rho = 1.41$,

$$\frac{x^2}{1-x} = 139.0070921986\dots, \qquad \frac{x^2}{\sqrt{1-x^2}} = 8.2924678943\dots.$$

The computation evaluates the Hardy scale and admissible dyadic Rudin–Shapiro lower examples at $\eta = 10^{-1}, 10^{-2}, 10^{-3}$, using lengths $8, 64, 512$. Non-dyadic lengths are rejected by the implementation because the displayed normalization is then not certified by the standard identity.

Remark 3.1. This theorem optimizes the conversion of a known norm into an absolute coefficient budget. It does not prove that the actual g_σ norm decays. Gates A–E remain false/open and no Hilbert–Polya or RH statement follows.